National Taipei University
Department of Communication Engineering
2024 Senior Project

國立臺北大學
National Taipei University

Incentive Design for Cooperative Cognitive Radio Networks

Advisor: *Yu-Pin Hsu* / **Presenter:** *Hung-Yi Lin*
Department of Communication Engineering

Introduction

As spectrum scarcity increasingly becomes a severe problem in the telecommunications industry, cognitive radio networks have been proposed to address the situation. However, incentivizing users to participate in the cooperative cognitive radio network scheme remains a challenge. The aim of this project is to explore incentive design for maximizing user satisfaction without involving transactions.

System Model

We consider the system sketch in Figure 1, where a primary (licensed) transmitter communicates with the intended receiver. In the same spectrum band, a secondary (unlicensed) composed of several transmitters-receivers pairs is seeking to exploit possible transmission opportunities.

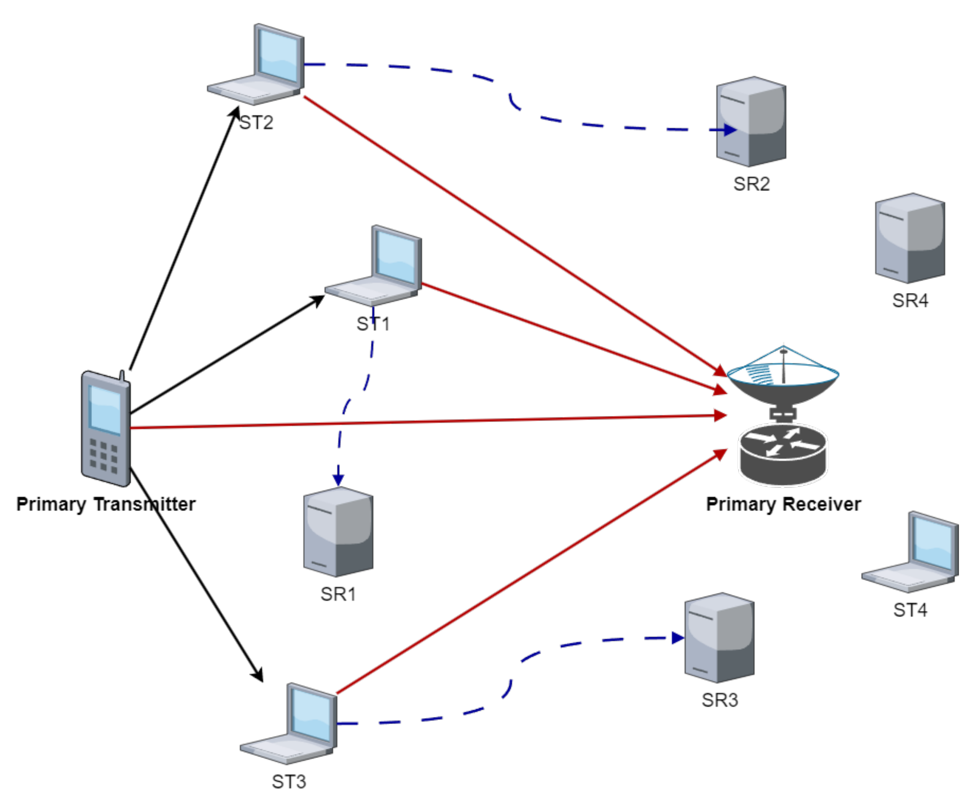

Figure 1: System model for cooperative cognitive networks

Problem Formulation

This is a cooperative cognitive radio network assuming that both users are rational and selfish, interested in maximizing their own utility. There are several parameters can be dynamically selected by each user.

Our problem focus on how to incentivize primary user to release it's transmission resources, specifically transmission time in this case.

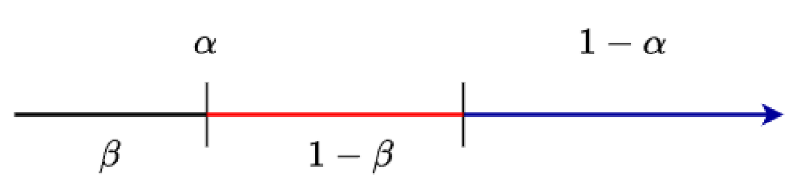

Figure 2: Duration of subslot corresponding to the color lines indicated in Figure 1

Solution Idea

Consider the entire model as a cooperative communication system. We apply local optimization techniques within the Stackelberg model, aiming to maximize the overall utilities of the entire system.

- Primary user's overall utility:

$$U_P = \frac{w_p}{1 + e^{-a(R_p(\alpha,\beta,P_s)-R_0)}}$$

- Secondary users' utility:

$$U_{S,i^{th}} = w_s(1-\alpha)R_{S,i} - \alpha(1-\beta)c_iP_s$$

w_p, w_s : The rate-weighted parameters
R_p : The overall achievable primary rate
R_0 : Primary user's traffic requirement

$R_{S,i}$: The transmission rate of i^{th} secondary link
c_i : Unit power transmitting costs
P_s : Secondary user's transmitting power

We aim to strike a balance between resource sharing and priority access under an incentive mechanism without payment. In other words, this research endeavors to find a Stackelberg equilibrium in such network system.

Simulation Results

- β^* increases when the normalized distance d becomes larger.
- There exists an β^* value for any given α under the same environment.
- The best response P_s^* of the secondary user decreases with increasing α.
- For primary user, P_s is not as attractive as α.

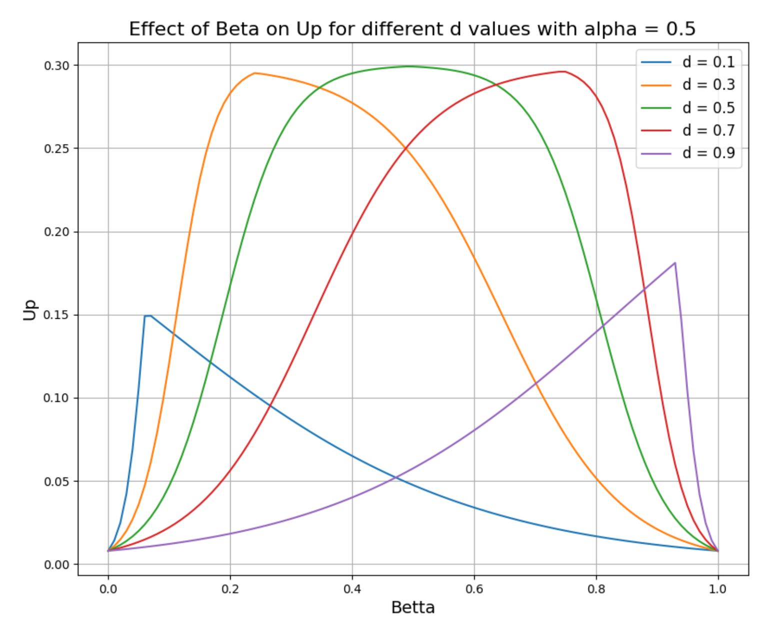

Figure 3: Optimal β for different d

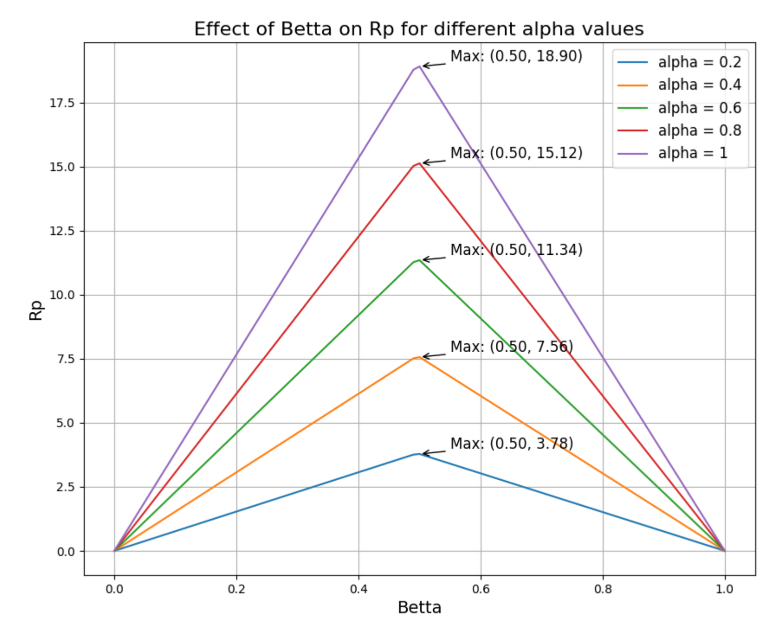

Figure 4: Optimal β for different α

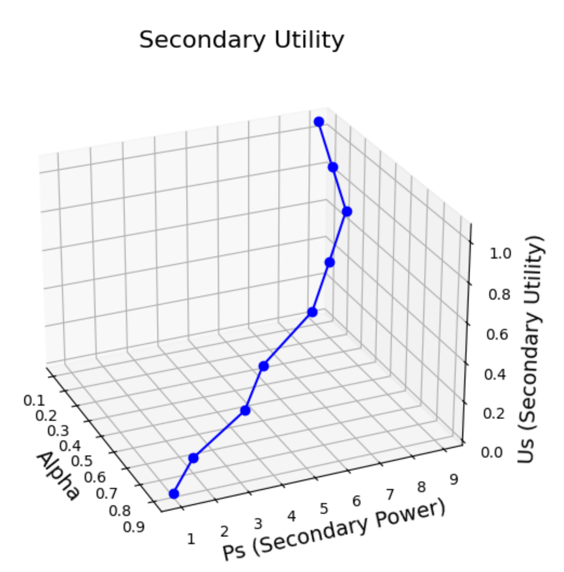

Figure 5: Secondary user's utility function

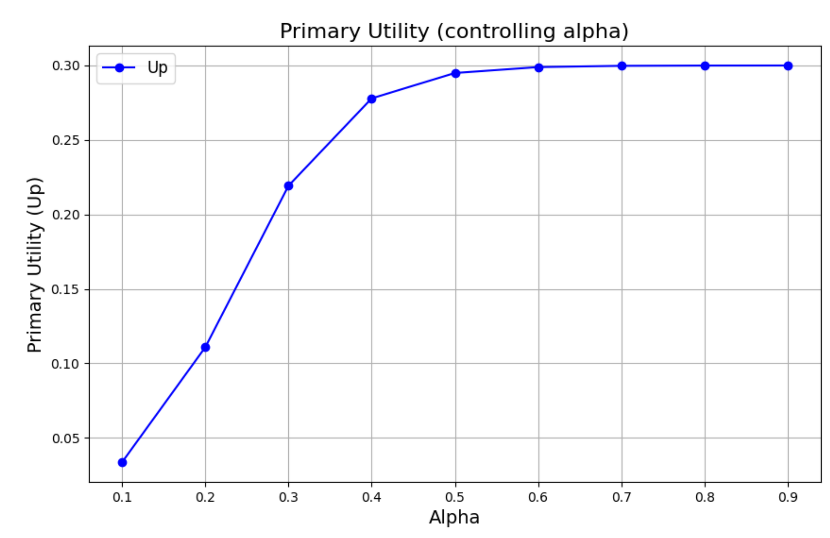

Figure 6: Primary user's utility function

Observations

1. Optimal P_s^* value:
   - Secondary user will control their transmitting power within a reasonable range, balancing between a high transmission rate and low cost.
2. Optimal α^* value:
   - There doesn't exist an optimal strategy α^* that satisfies both users.
   - The primary user is less inclined to share the transmission time, as the influence of the secondary user's transmitting power is not deemed significant from the primary user's perspective.

Conclusions

- The ideal Stackelberg equilibrium doesn't exist in this system. Without monetary transactions, dividing transmission time among secondary users isn't practical for primary user.
- Mathematically, this outcome can be attributed to the contrasting growth rates between linear and logarithmic functions.

References

[1] Zhang, J., & Zhang, Q. "Stackelberg game for utility-based cooperative cognitiveradio networks." *Proceedings of the tenth ACM international symposium on Mobile ad hoc networking and computing.* 2009.

[2] Michael Maschler, Eilon Solan, and Shmuel Zamir. *"Game Theory" 2nd edition* : Cambridge University Press 2020 ISBN-10:110849